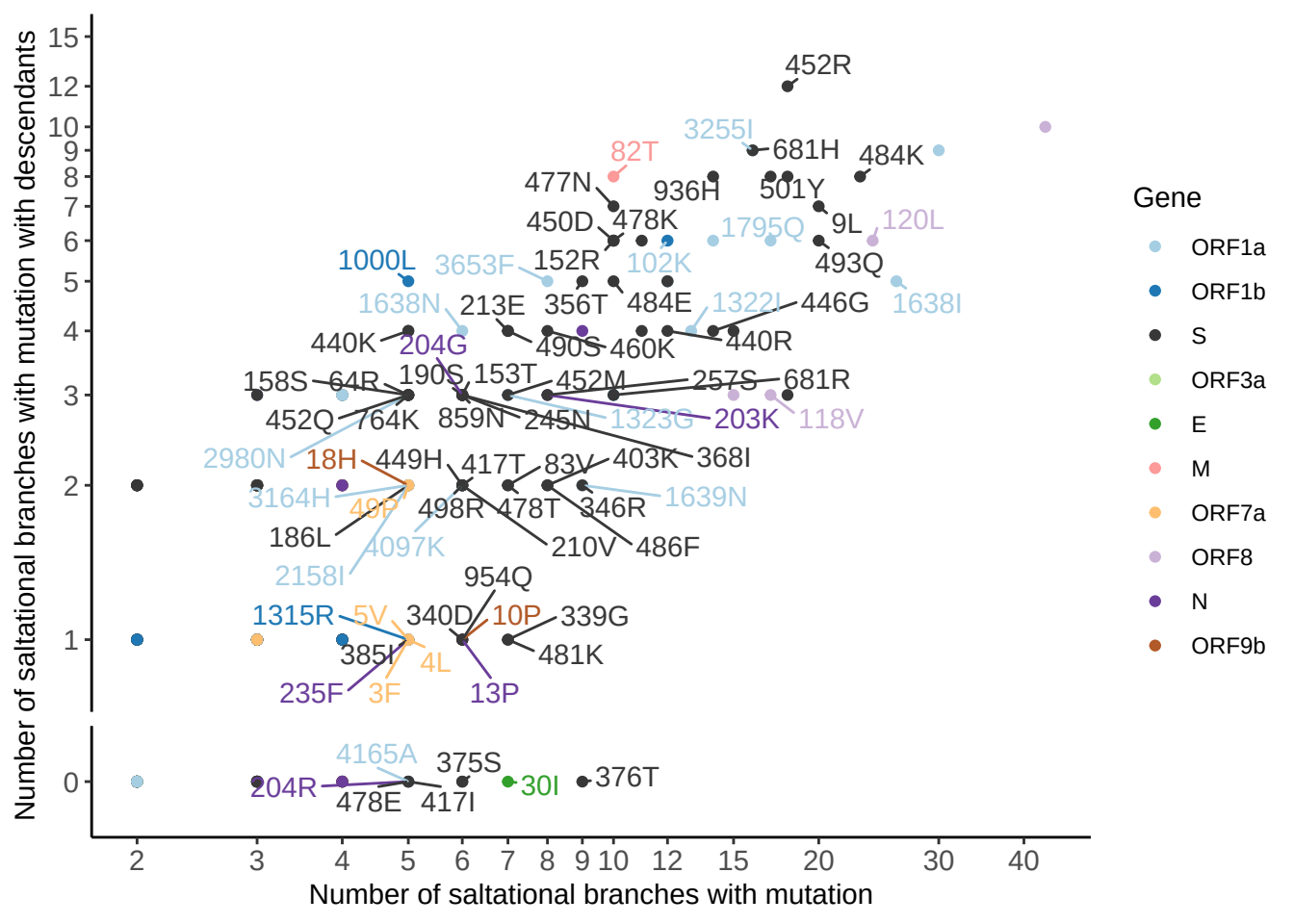
Number of saltational branches with mutation with descendants
Number of saltational branches with mutation
Gene
ORF1a
ORF1b
S
ORF3a
E
M
ORF7a
ORF8
N
ORF9b
452R
3255I
82T
681H
484K
477N
936H
501Y
450D
478K
1795Q
9L
120L
1000L
3653F
152R
102K
493Q
1638N
213E
356T
484E
1322L
446G
1638I
440K
204G
490S
460K
440R
158S
64R
190S
153T
452M
257S
681R
452Q
764K
859N
245N
1323G
203K
118V
2980N
18H
449H
417T
83V
403K
368I
3164H
49P
498R
478T
346R
1639N
186L
4097K
210V
486F
2158I
954Q
1315R
5V
340D
10P
339G
385I
4L
481K
235F
3F
13P
4165A
375S
204R
478E
417I
30I
376T